PAULA MAGDY

Full Stack Software Engineer

(+20) 127 776 7028 • paulamagdy665@gmail.com • Portfolio • GitHub • LinkedIn • Cairo, Egypt

PROFILE

Full Stack Software Engineer with 3 years of professional experience building production web applications with React, Next.js, TypeScript, and Node.js/NestJS, backed by PostgreSQL and MySQL. Experienced in REST APIs, authentication and access control, automated testing, CI/CD, and AI-assisted development on AWS and Azure. AWS Certified Solutions Architect.

SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, PHP, SQL, C++

Frontend: React.js (Hooks, Context, Redux), Next.js, React Native, HTML5, CSS3, Tailwind CSS, Shadcn/UI, Bootstrap

Backend & APIs: Node.js, NestJS, Express.js, Laravel, Django, Flask, REST APIs, Swagger/OpenAPI, WebSockets, JWT, OAuth, RBAC

Architecture: Microservices, message queues (Amazon SQS), webhooks, background jobs, LLM APIs, Paymob, Twilio

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Prisma ORM, query optimization

Cloud: AWS (EC2, Lambda, API Gateway, S3, RDS, DynamoDB, ECS, CloudFormation), Azure, Huawei Cloud

DevOps: Docker, Kubernetes, CI/CD (GitHub Actions, GitLab CI, CodePipeline), Linux, Nginx

Testing & Practices: Jest, Vitest, React Testing Library, Playwright, PHPUnit, Pest, TDD, Git, Domain-Driven Design, Agile/Scrum

PROFESSIONAL EXPERIENCE

Challenge Group | Full-Stack Software Engineer **Jun 2025 – Present**

Cairo, Egypt · Production platforms for government and enterprise clients across Egypt, KSA, and the UAE

– Developed domain-driven NestJS modules and Laravel APIs using PostgreSQL/Prisma and MySQL for production platforms serving 5,000+ concurrent users; reduced API latency by approximately 35% through targeted indexing and query tuning.

– Built and maintained Next.js/React interfaces, responsive shared components, role-based admin panels, onboarding and results-entry flows, and live-updating leaderboards.

– Implemented secure API patterns with Sanctum/JWT authentication, role-based access control, request validation, and access restrictions to reject unauthorized or malformed requests.

– Maintained automated tests using Vitest, Jest, Pest, and PHPUnit in CI workflows to help catch regressions before release.

– Supported container-based CI/CD with Docker, ECR, ECS, CodeBuild, and CodePipeline, reducing deployment time from approximately 20 minutes to under 5.

– Integrated AI/LLM APIs and third-party and real-time services including Paymob payments, Twilio voice/SMS, FCM push notifications, S3 media storage, and Laravel Reverb WebSockets.

– Built backend services using microservices architecture and message queues (Amazon SQS).

– Provisioned AWS infrastructure including VPC, EC2, RDS, S3, CloudFront, IAM, and Secrets Manager; monitored services with CloudWatch and structured logging.

– Independently designed and shipped an Amazon Lex chatbot on AWS Lambda integrated with Twilio to automate customer-support conversations.

– Used AI-assisted workflows for planning, coding, and testing, took part in code reviews, mentored developers, and wrote technical documentation.

Stack: *React, Next.js, TypeScript, Node.js, NestJS, Laravel, PostgreSQL, MySQL, Redis, Docker, AWS, Azure*

Information Technology Institute (ITI) | Full-Stack Engineer **Oct 2024 – Jun 2025**

Cairo, Egypt · Government-sponsored intensive software-engineering program (~1,200 hours)

– Built full-stack applications using Node.js/Express, Python (Django, Flask), and PHP/Laravel with React frontends, secure REST APIs, structured authentication, and role-based access control.

– Modeled relational SQL and NoSQL databases, applying normalization and indexing to support efficient queries.

– Containerized services with Docker and deployed to AWS while applying system-design fundamentals and Agile delivery practices.

Freelance | Front-End Developer **Oct 2023 – Oct 2024**

Cairo, Egypt · Client-facing frontend development, primarily for e-commerce websites

– Built and shipped responsive React storefronts connected to REST APIs for product catalogs and cart functionality.

– Created reusable components and centralized catalog/cart state with React Context API, improving maintainability and consistency.

– Optimized rendering and asset loading to improve page-load performance and keep large product listings responsive.

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (SAA-C03) **Jun 2026**

Design of fault-tolerant, scalable, and secure AWS architectures, including networking, compute, storage, databases, IAM, and high availability.

AWS Certified Cloud Practitioner **Apr 2026**

Cloud fundamentals, shared-responsibility security, pricing, billing, and core AWS services.

Huawei HCCDP – Solution Architectures **Jun 2026**

End-to-end solution design on Huawei Cloud with focus on high availability, security, and cost efficiency.

Huawei HCCDA – Tech Essentials **May 2026**

Foundational Huawei Cloud knowledge covering compute, storage, networking, deployment, and developer tooling.

EDUCATION

B.Sc., Computer Science and Information Systems **2018 – 2022**

October 6 University, Giza, Egypt

– Relevant coursework: Algorithms, Data Structures, Cloud Computing, Database Systems, Software Engineering.

– Graduation project (Grade A): Learning Management System built with Node.js, Express, and MongoDB.